

INTERNAL

Operator/country market share by count, mass, slot value; HHI concentration.

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DATASETS

space/space_track_satcat, space/operator_mapping

KEY FINDINGS

- Top operator by count is ISRO with 225 satellites (7.04% share).
- Operator HHI by count = 333.6; by mass = 336.8.
- Top country by count is United States with 48.95% share.

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Data Sources

- **space/space_track_satcat**
- **space/operator_mapping**

Key Findings

- Top operator by count is ISRO with 225 satellites (7.04% share).
- Operator HHI by count = 333.6; by mass = 336.8.
- Top country by count is United States with 48.95% share.

Executive Summary

NOTE

Measures operator and country concentration in the satellite catalog using satellite count, launch mass, and GEO slot occupancy. HHI quantifies market concentration across each dimension.

Results & Interpretation

NOTE

Operator/country market share by count, mass, slot value; HHI concentration.

Main takeaways:

- Top operator by count is ISRO with 225 satellites (7.04% share).
- Operator HHI by count = 333.6; by mass = 336.8.
- Top country by count is United States with 48.95% share.

Objectives

- Rank operators and countries by active satellite count and mass.
- Quantify concentration with the Herfindahl-Hirschman Index.
- Assess GEO orbital-slot ownership as a proxy for slot value.

Methodology

NOTE

SATCAT records are loaded from the space-track dataset, enriched with derived orbital columns, and aggregated by operator and country. Shares are computed as each group's count or mass divided by the total. HHI is the sum of squared share percentages.

Charts

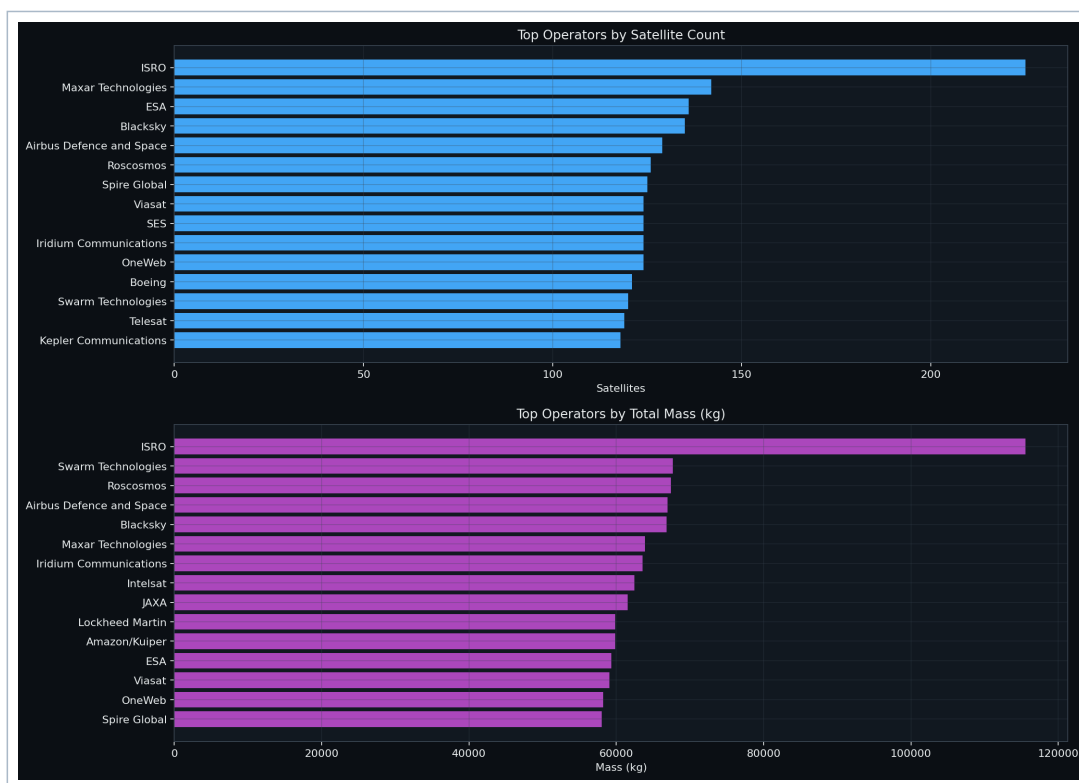


Figure 1. Operator share of sky
Operator share of sky

Statistical Results

operator	count_or_wei...	share_pct
ISRO	225	+ 7.04%
Maxar Technologies	142	+ 4.44%
ESA	136	+ 4.26%
Blacksky	135	+ 4.23%
Airbus Defence and Space	129	+ 4.04%
Roscosmos	126	+ 3.94%
Spire Global	125	+ 3.91%
Viasat	124	+ 3.88%
SES	124	+ 3.88%
Iridium Communications	124	+ 3.88%
OneWeb	124	+ 3.88%
Boeing	121	+ 3.79%
Swarm Technologies	120	+ 3.76%
Telesat	119	+ 3.72%
Kepler Communications	118	+ 3.69%

operator	count_or_wei...	share_pct
JAXA	118	+ 3.69%
Lockheed Martin	117	+ 3.66%
Northrop Grumman	116	+ 3.63%
Intelsat	114	+ 3.57%
Planet Labs	111	+ 3.47%

Table 1. Operator share by satellite count

operator	count_or_wei...	share_pct
ISRO	115,539	+ 7.31%
Swarm Technologies	67,688	+ 4.28%
Roscosmos	67,438	+ 4.27%
Airbus Defence and Space	66,958	+ 4.24%
Blacksky	66,838	+ 4.23%
Maxar Technologies	63,886	+ 4.04%
Iridium Communications	63,540	+ 4.02%
Intelsat	62,483	+ 3.95%
JAXA	61,571	+ 3.90%
Lockheed Martin	59,878	+ 3.79%
Amazon/Kuiper	59,816	+ 3.79%
ESA	59,322	+ 3.75%
Viasat	59,035	+ 3.74%
OneWeb	58,234	+ 3.69%
Spire Global	57,995	+ 3.67%
Boeing	57,853	+ 3.66%
Eutelsat	56,725	+ 3.59%
Northrop Grumman	56,204	+ 3.56%
Kepler Communications	56,023	+ 3.55%
SpaceX	55,173	+ 3.49%

Table 2. Operator share by total mass

country	count_or_wei...	share_pct
United States	1,564	+ 48.95%
Canada	335	+ 10.49%
Europe	265	+ 8.29%

country	count_or_wei...	share_pct
India	225	+ 7.04%
Russia	126	+ 3.94%
Luxembourg	124	+ 3.88%
United Kingdom	124	+ 3.88%
Japan	118	+ 3.69%
France	110	+ 3.44%
France/Italy	102	+ 3.19%
China	102	+ 3.19%

Table 3. Country share by satellite count

metric	value
HHI by count	333.60
HHI by mass	336.76
GEO objects	0
Total objects	3,195

Table 4. Concentration metrics

Limitations

NOTE

Results are research-grade, not investment advice. The sample is limited to available data, and measured edges may not persist out-of-sample.

Appendix Tables

concentration_metrics

metric	value
HHI by count	333.60
HHI by mass	336.76
GEO objects	0
Total objects	3,195

country_share_count

country	count_or_wei...	share_pct
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operator_share_mass

operator	count_or_wei...	share_pct
ISRO	115,539	+ 7.31%
Swarm Technologies	67,688	+ 4.28%

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Statistics Summary

Key	Value
study	operator_share_of_sky
category	alternative_data
status	success
parameters	{}
datasets	['space/space_track_satcat', 'space/operator_mapping']
rows	0
git_commit	e97f0b72579223cba60ddd3e4702b560febfe339
executed_at	2026-08-04T12:26:56.430878+00:00

Glossary

Term	Definition
n / count	Number of observations or events in the sample.
edge	Average excess return versus the unconditional baseline.
hit-rate	Share of events with a positive forward return.
p-value	Probability of observing the result under the null of no edge.
*** / ** / *	Significance at 99.9% / 99% / 95% confidence.
α (alpha)	Intercept or active return versus benchmark.

Term	Definition
β (beta)	Sensitivity to a reference factor or market.
σ (sigma)	Volatility / standard deviation of returns.

Document Control

Field	Value
Report schema	2
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Study slug	operator_share_of_sky
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Git commit	e97f0b72
Datasets	space/space_track_satcat, space/operator_mapping
Data fingerprint	c1c6bb9ba575
Duration	0.2s

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